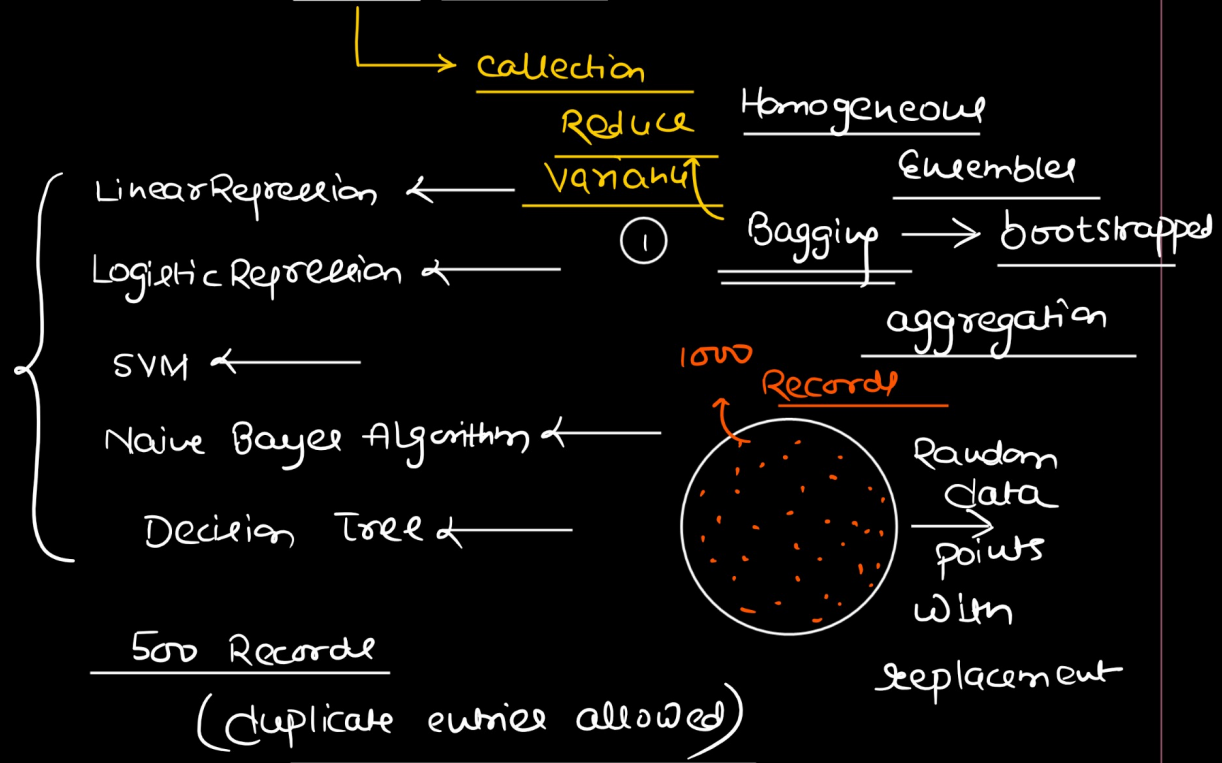


Ensemble Models



breast_cancer ← Decision Tree

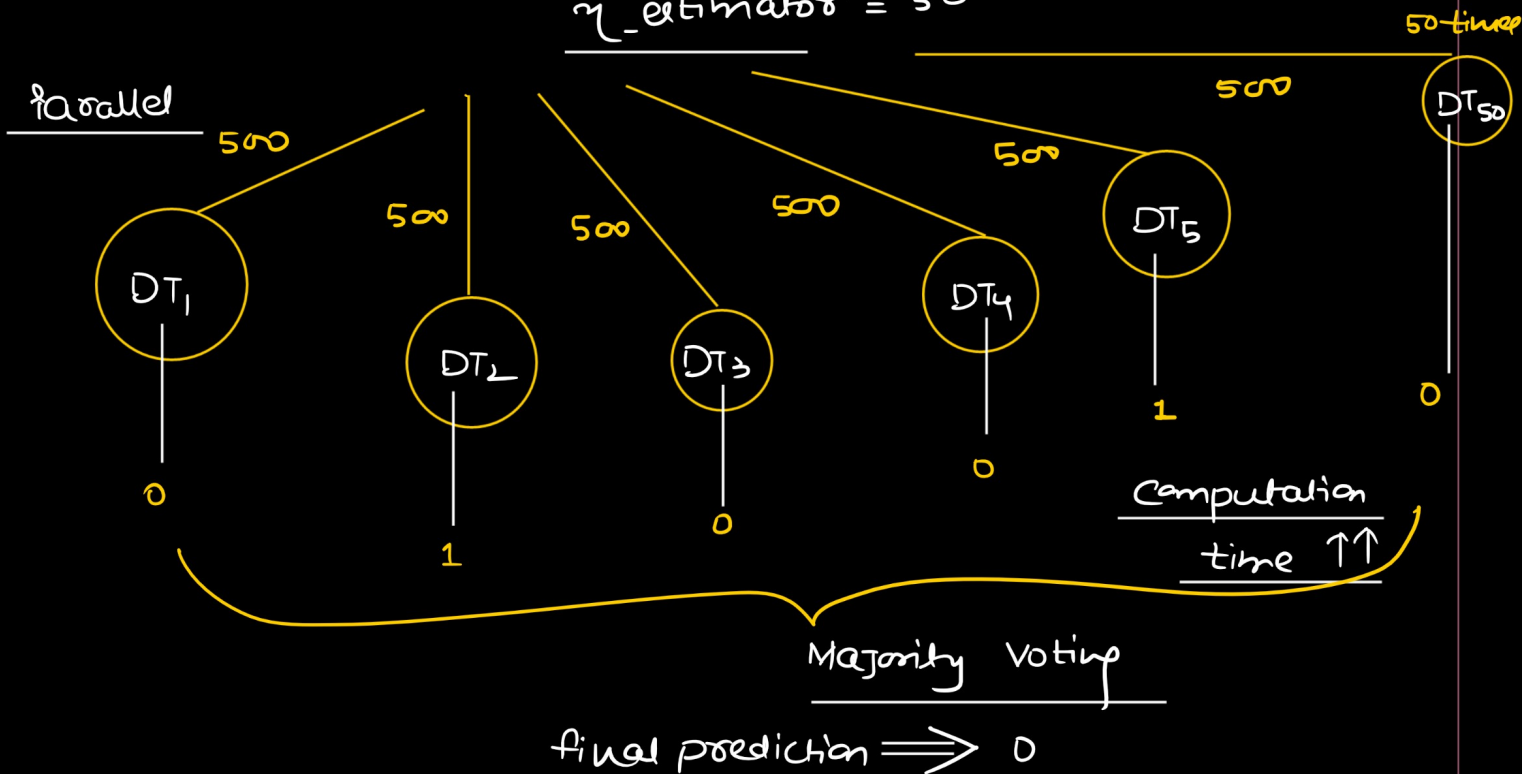
→ overfitting

Randomforest ← Bagging classifier

(low bias, high variance)

test-set → Patient input features

$n_{estimator} = 50$

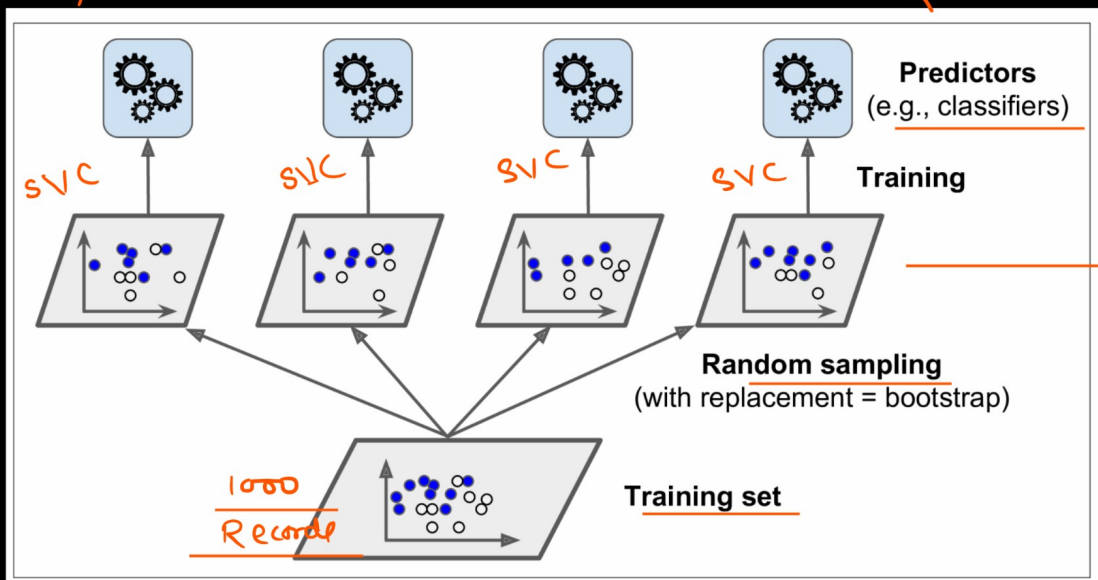
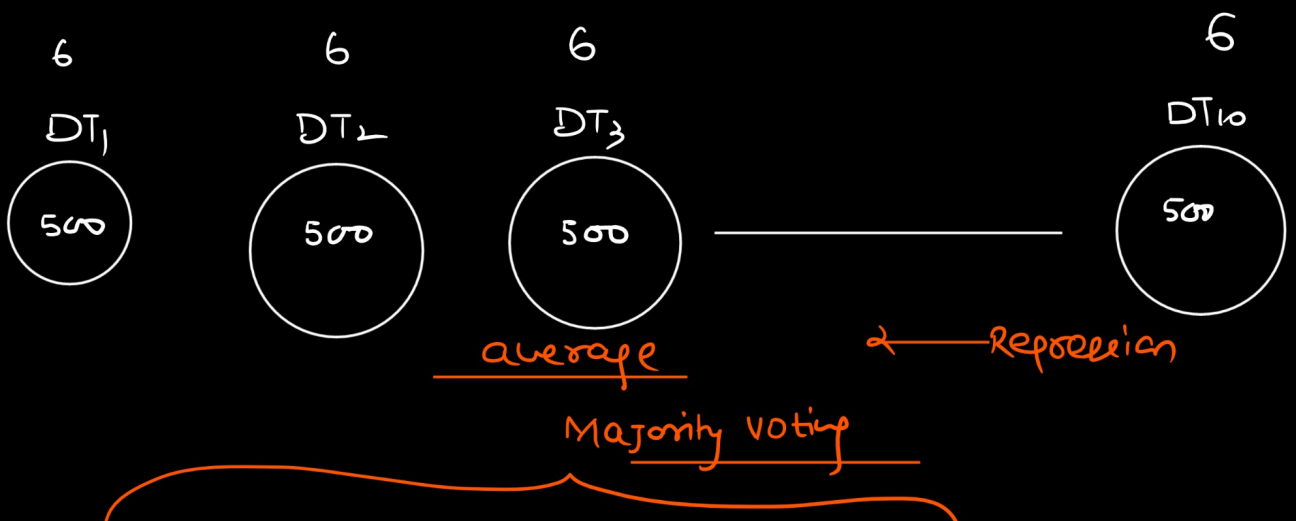


overfitting ↓ ↓

Randomly select the records with replacement

	f_1	f_2	f_3	f_4	f_5	f_6	
0							(1000, 6)
1							
2							
3							
999							

bootstrap - feature = false



Pruning → Can avoid the issue of overfitting but with limited
max_depth → underfitting of high bias, extra
low
variance

original data

→ [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

bootstrap sample

→ [2, 3, 3, 5, 7, 8, 8, 10, 1, 2]

Random sample
with
replacement

Misfit records

test set
to evaluate
the

→ OOB → out of bag sample

performance
of the model

records not selected
in the bootstrap

sample → OOB sample

for any observation, probability of being
selected in one draw

$\frac{1}{n}$ → size of
the
record

Probability of not being selected in one draw
 $\Rightarrow \left(1 - \frac{1}{n}\right)$

Probability of not being selected in n draw

OOB $\Rightarrow$

$$\Rightarrow \left(1 - \frac{1}{n}\right)^n$$

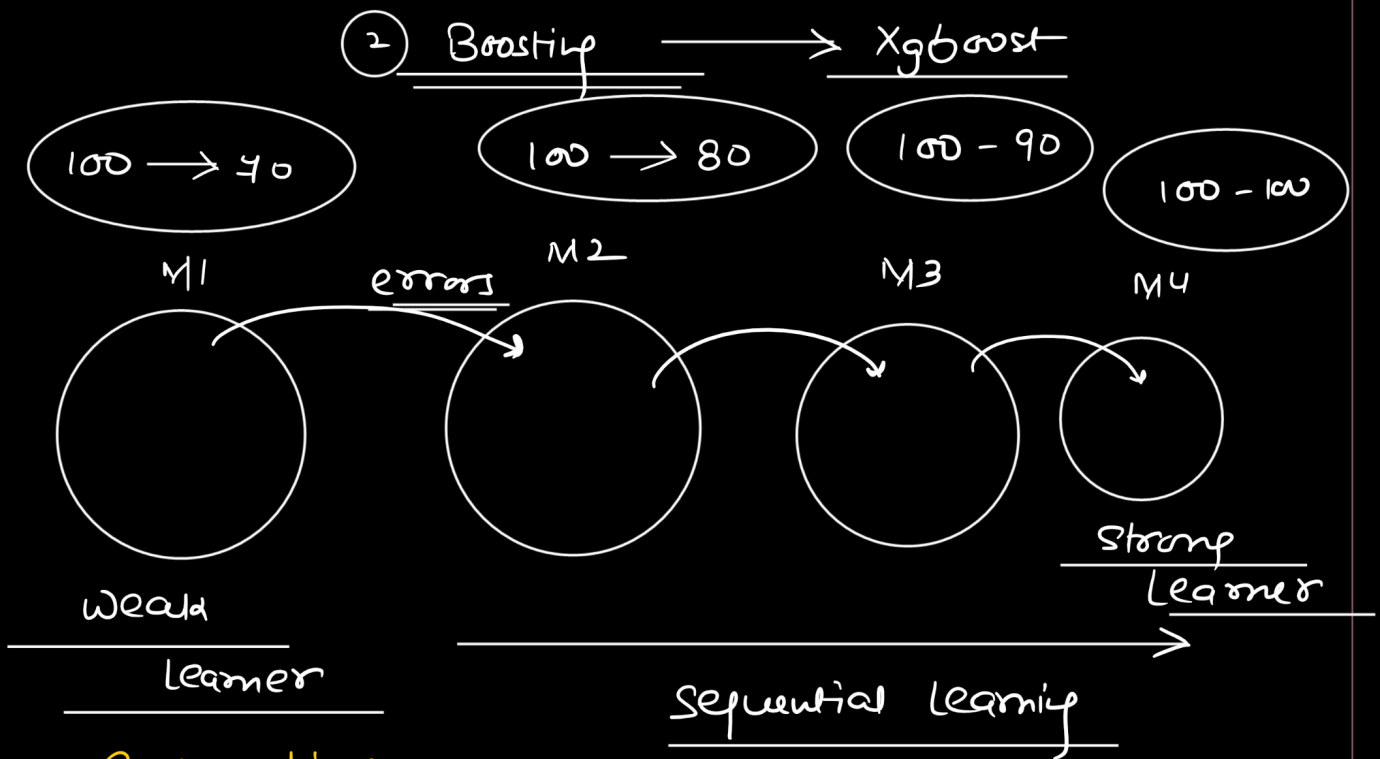
n is very very large

$$\approx e^{-1} \approx 0.368$$

36.8% data	$\rightarrow$ <u>OOB</u>
63.2% data	$\rightarrow$ <u>gets selected</u>

Final Conclusion

Bagging (Bootstrap Aggregating) is an ensemble machine learning technique designed to reduce variance and prevent overfitting. It works by training multiple base models independently in parallel, using random subsets of the dataset, and then combining their predictions through averaging or majority voting.

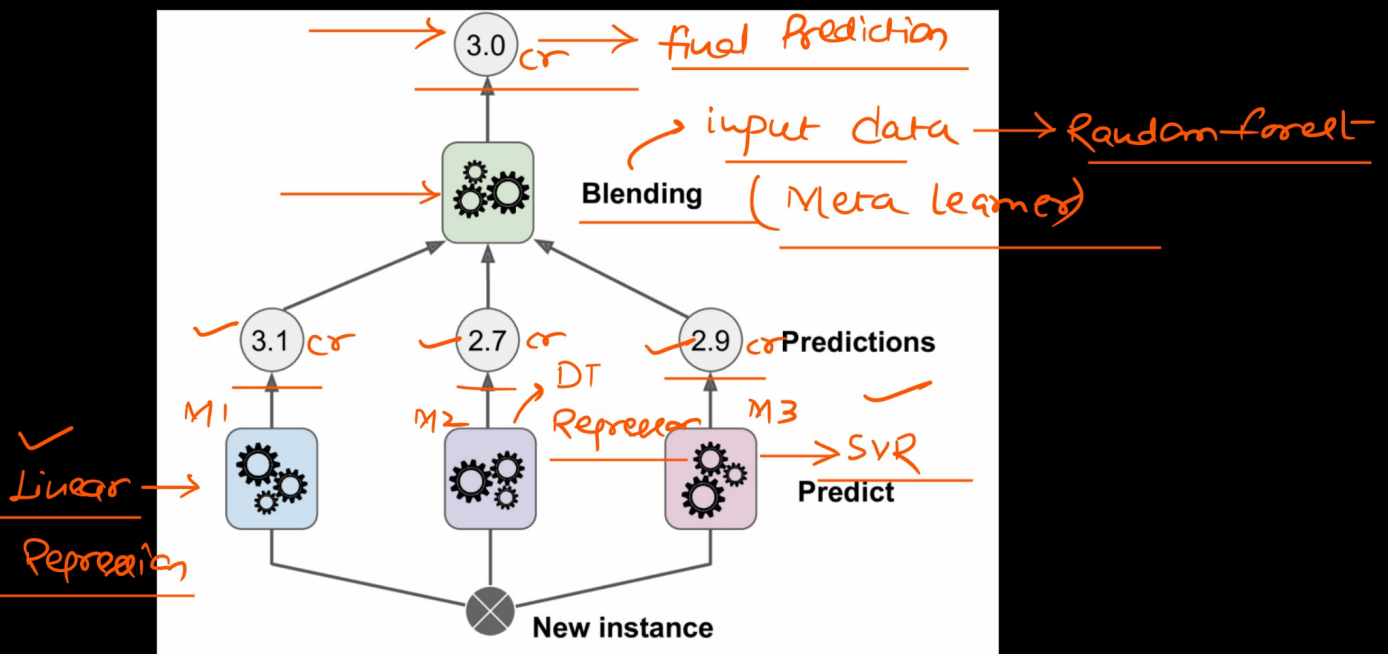


→ Reduce bias

Boosting is an ensemble learning technique that improves a model's predictive accuracy by combining multiple "weak learners" into a single "strong learner".

1. Models are trained one after another
2. Misclassified samples receive higher importance
3. Subsequent learners focus more on difficult instances
4. Final prediction is a weighted combination of all learners

③ Stacking → Heterogeneous ensemble model



Randomforest

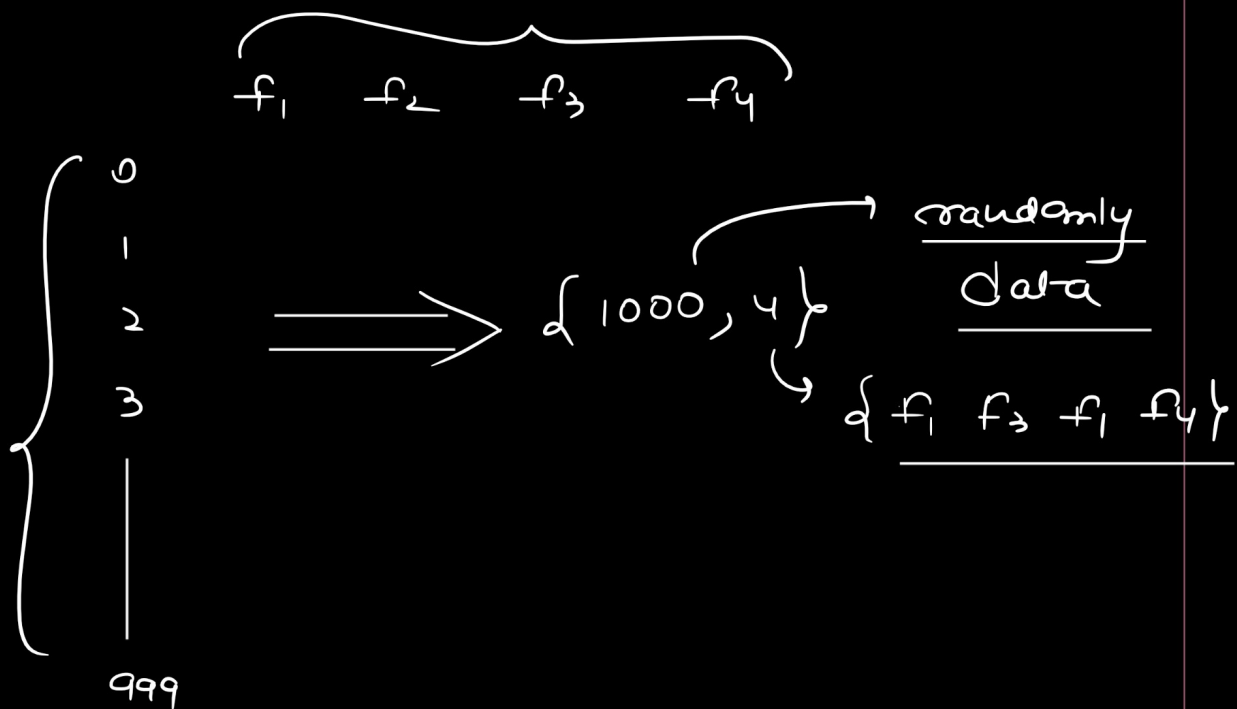
→ bagging

↳ DecisionTree (collection of multiple
Decision trees)

Bootstrap sampling + Random feature selection

↳ to have a random records &
features

With replacement



EmpID, Name, Salary, Experience, Education, City

Bagging classifier →

D1 → DT1

D2 → DT2

D3 → DT3

D4 → DT4

Contains all
the
features

Randomforest →

D1 → DT1 (Age, salary,
city)

D2 → DT2 (Experience,
city, Education)

D3 → DT3 (Age, Education,
Salary)

More

Variation

are

comparable to the

above models

unique feature